GEST-S2701 – Exercise Session 8

ES 8. Leasing and Financial debts

2025-2026

Solvay Brussels School Economics & Management

Exercise 8.1

A transport and logistics company, S&L, needs to acquire a new delivery truck.
To do this, she considers the following three scenarios .

Note: For this entire exercise, we assume that there is no VAT.

Scenario A: On 01/08/2023, the company S&L purchases a new truck worth €52,000. The invoice is paid in full by bank transfer on 08/15/2023. Furthermore, the purchase is financed entirely by a loan with constant annuities and annual payments, for a term of 5 years and with an annual interest rate of 4% (payable at the end of the term).

Exercise 8.1 – Scenario A

1) Record the transactions in the journal and identify the direct and indirect impacts on cash flows:

   a) At the time of purchase of the truck.

Exercise 8.1 – Scenario A – 1a) Purchase of the vehicle

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
01/08/2023	24.0		Furniture and vehicles: Acquisition value	52,000	
		440	Suppliers		52,000
15/08/2023	440		Suppliers	52,000	
		550.0	Credit institutions: Current account		52,000
	550.0		Credit institutions: Current account	52,000	
		1730	Credit institutions: Current account payable		52,000

Exercise 8.1 – Scenario A – 1a) Purchase of the vehicle

- Impact on cash flows – 01/08/2023

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	0	0	0	0	0
Total impact	**0**	**0**	**0**	**0**	**0**

Exercise 8.1 – Scenario A – 1a) Purchase of the vehicle

- Impact on cash flows – 15/08/2023

Timing	**CFop (1)**	**CFinv (2)**	**FCF (3) = (1) + (2)**	**CFfin (4)**	**Δ CASH (5) = (3) + (4)**
Direct impact: at accounting entry	0	-52,000	-52,000	+52,000	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	0	0	0	0	0
Total impact	**0**	**-52,000**	**-52,000**	**+52,000**	**0**

Exercise 8.1 – Scenario A

1) Record the transactions in the journal and identify the direct and indirect impacts on cash flows :

    b) At the close of the first financial year, on 31/12/2023, knowing that the truck has an estimated lifespan of 5 years.

Exercise 8.1 – Scenario A – 1b) Closing of the books

- Accouting journal - depreciation

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/12/2023	6302		Depreciation of tangible fixed assets	4,333.33	
		24.9	Furniture and vehicles: Depreciation recorded		4,333.33

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Exercise 8.1 – Scenario A – 1b) End of first FY

Loan amount (V)	52.000,00
Loan term (T)	5
Annual interest rate (r)	4%
Annuity	€ 11 680,61

$$Annuity = \frac{V \times r}{1 - \frac{1}{(1+r)^T}} = \frac{52{,}000 \times 0.04}{1 - \frac{1}{(1+0.04)^5}}$$

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+52,000.00			52,000.00
1	01/08/24	-11,680.61	$\boxed{\text{-2,080.00}}$	$\boxed{\text{-9,600.61}}$	42,399.39
2	01/08/25	-11,680.61	-1,695.98	-9,984.63	32,414.76
3	01/08/26	-11,680.61	-1,296.59	-10,384.02	22,030.74
4	01/08/27	-11,680.61	-881.23	-10,799.38	11,231.36
5	01/08/28	-11,680.61	-449.25	-11,231.36	-

$$= -r * OA_{t-1}$$
$$= -4\% * 52.000{,}00$$

$$= -(Annuity - Interests)$$
$$= -(11.680{,}81 - 2.080{,}00)$$

Exercise 8.1 – Scenario A – 1b) End of first FY

- Accounting journal – Accrued interest + reclassification of debts

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/12/2023	6500		Interest, commission and other charges relating to debts	866.67	
		492	Accrued charges		866/67
31/12/2023	1730		Credit institutions: Current account payable	9,600.61	
		4230	Current portion of amounts payable after more than one year falling due within one year		9,600.61

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+1.300	0	+1.300	0	+1.300
Total impact	**+1.300**	**0**	**+1.300**	**0**	**+1.300**

= (4,333.33 + 866.67) * 25%

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Exercise 8.1 – Scenario A

1) Record the transactions in the journal and identify the direct and indirect impacts on cash flows:

    c) During the second financial year (excluding closing operations).

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Exercise 8.1 –Scenario A – 1c) Accruals

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
01/01/2024	492		Accrued charges	866.67	
		6500	Interest, commission and other charges relating to debts		866.67

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-216.67	0	-216.67	0	-216.67
Total impact	**-216.67**	**0**	**-216.67**	**0**	**-216.67**

Exercise 8.1 – Scenario A – 1c) Annuity payment

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/07/2024	4230		Current portion of amounts payable after more than one year falling due within one year	9,600.61	
	6500		Interest, commission and other charges relating to debts	2,080	
		550.0	Credit institutions: Current account		11,680.61

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	-2,080.00	0	-2,080.00	-9,600.61	-11,680.61
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+520	0	+520	0	+520
Total impact	**-1,560**	**0**	**-1,560**	**-9,600.61**	**-11,160.61**

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Exercise 8.1

Scenario B : Le On 01/08/2023, the company S&L signs a financial lease agreement for the acquisition of a new truck, with the following characteristics:

- Vehicle value : €52,000
- Contract duration : 5 years
- Interest rate (payable in arrears) : 4%
- Payment frequency : annual
- Purchase option at the end of the contract : €6,000

Exercise 8.1 – Scenario B

2) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L :

   a) At the beginning of the contract.

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Exercise 8.1 – Scenario B – 2a) Begininning of the contract

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
01/08/2023	252.0		Fixed assets in leasing or other similar rights: Furniture and vehicles: Acquisition value	46,000	
		172	Leasing and other similar obligations		46,000

- Impact on cash flows

Timing	CFop (1)	Cfinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	-46,000	-46,000	+46,000	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	0	0	0	0	0
Total impact	**0**	**-46,000**	**-46,000**	**+46,000**	**0**

Exercise 8.1 – Scenario B – 2b) End of first FY

2) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L:

    b) At the end of the first financial year.

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Exercise 8.1 – Scenario B – 2b) End of first FY

- Accounting journal - Depreciation

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/12/2023	6302		Depreciation of tangible fixed assets	3.833,33	
		252.9	Fixed assets in leasing or other similar rights: Furniture and vehicles: Depreciation recorded		3.833,33

Exercise 8.1 – Scenario B – 2b) End of first FY

Vehicle value	52.000,00
Purchase option at the end of the contract	6.000,00
Contract value (V)	**46.000,00**
Contract duration (T)	5
Annual interest rate (r)	4%
Annuity	€ 10 332,85

$$Annuity = \frac{V \times r}{1 - \frac{1}{(1+r)^T}} = \frac{46{,}000 \times 0.04}{1 - \frac{1}{(1+0.04)^5}}$$

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+46,000.00			46,000.00
1	01/08/24	-10,332.85	-1,840.00	-8,492.85	37,507.15
2	01/08/25	-10,332.85	-1,500.29	-8,832.56	28,674.59
3	01/08/26	-10,332.85	-1,146.98	-9,185.87	19,488.72
4	01/08/27	-10,332.85	-779.55	-9,553.30	9,935.42
5	01/08/28	-10,332.85	-397.42	-9,935.43	-

$$= -r * OA_{t-1}$$
$$= -4\% * 46.000,00$$

$$= -(Annuity - Interests)$$
$$= -(10.332,85 - 1.840,00)$$

Exercise 8.1 – Scenario B – 2b) Closing of first FY

- Accounting journal – Accrued interest and debt reclassification

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/12/2023	6500		Interest, commission and other charges relating to debts	766.67	
		492	Accrued charges		766.67
31/12/2023	172		Leasing and other similar obligations	8,492.85	
		422	Leasing and other similar obligations [short term]		8,492.85

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+1,150	0	+1,150	0	+1,150
Total impact	**+1,150**	**0**	**+1,150**	**0**	**+1,150**

= (3,833.33 + 766.67) * 25%

Exercise 8.1 – Scenario B

2) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L:

   c) During the second financial year (excluding closing operations).

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Exercise 8.1 – Scenario B – 2c) Accruals

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
01/01/2024	492		Accrued charges	766.67	
		6500	Interest, commission and other charges relating to debts		766.67

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-191.67	0	-191.67	0	-191.67
Total impact	**-191.67**	**0**	**-191.67**	**0**	**--191.67**

Exercise 8.1 – Scenario B – 2c) Annuity payment

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/07/2024	422		Leasing and other similar obligations [short term]	8,492.85	
	6500		Interest, commission and other charges relating to debts	1,840.00	
		550.0	Credit institutions: Current account		10,332.85

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	-1,840	0	-1,840	-8,492.85	-10,332.85
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+460	0	+460	0	+460
Total impact	**-1,380**	**0**	**-1,380**	**-8,492.85**	**-9,872.85**

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Exercise 8.1 – Scenario B

2) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L:

   d) At the expiration of the contract in the event that the tenant exercises the purchase option.

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Exercise 8.1 – Scenario B – 2d) Option is paid

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/07/2028	422		Leasing and other similar obligations [short term]	9,935.43	
	6500		Interest, commission and other charges relating to debts	397.42	
		550.0	Credit institutions: Current account		10,332.85
31/07/2028	24.0		Mobilier et matériel roulant : Valeur d'acquisition	6,000	
		440	Suppliers		6,000
	440		Suppliers	6,000	
		550.0	Credit institutions: Current account		6,000
	252.9		Fixed assets in leasing or other similar rights: Furniture and vehicles: Depreciation recorded	40,633.33	
	24.0		Furniture and vehicles: Acquisition value	46,000	
		24.9	Furniture and vehicles: Depreciation recorded		40,633.33
		252.0	Fixed assets in leasing or other similar rights: Furniture and vehicles: Acquisition value		46,000

NB : The last recording of the depreciation is on le 31/12/2028.

Exercise 8.1 – Scenario B – 2d) Option is paid

- Impact on cash flows

Timing	CFop (1)	Cfinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	-397,42	-6.000	-6.397,42	-9.935,43	-16.332,85
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+ 99,35	0	+99,35	0	+99,35
Total impact	**-298,07**	**-6.000**	**-6.298,07**	**-9.935,43**	**-16.233,50**

Exercise 8.1 – Scenario B

2) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L:

    e) At the expiration of the contract in the event that the tenant does not exercise the purchase option.

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Exercise 8.1 – Scenario B – 2e) Option is not paid

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/07/2028	422		Leasing and other similar obligations [short term]	9,935.43	
	6500		Interest, commission and other charges relating to debts	397.42	
		550.0	Credit institutions: Current account		10,332.85
31/07/2028	6302		Depreciation of tangible fixed assets	5,366.67	
		252.9	Fixed assets in leasing or other similar rights: Furniture and vehicles: Depreciation recorded		5,366.67
	252.9		Fixed assets in leasing or other similar rights: Furniture and vehicles: Depreciation recorded	46,000	
		252.0	Fixed assets in leasing or other similar rights: Furniture and vehicles: Acquisition value		46,000

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Exercise 8.1 – Scenario B – 2e) Option is not paid

- Impact on cash flows

Timing	CFop (1)	Cfinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	-397.42	0	-397.42	-9,935.43	-10,332.85
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+1,441.02	0	+1,441.02	0	+1,441.02
Total impact	**+1,043.60**	**0**	**+1,043.60**	**-9,935.43**	**-8,891.83**

= (397.42 + 5,366.67) * 25%

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Exercise 8.1 – Scenario C

Scenario C : On 1 August 2023, S&L entered into an operating lease agreement for a new truck :

- Vehicle value: €52,000
- Term of the agreement: 5 years
- Fixed annual lease payment (including insurance and road tax) of €10,800, payable in arrears.

Exercise 8.1 – Scenario C

4) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L :

   a) At the beginning of the contract.

Exercise 8.1 – Scenario C – 4a) Beginning of the contract

→ Nothing to be recorded in the accounting journal!

Exercise 8.1 – Scenario C

4) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L:

   b) At the end of the first financial year

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Exercise 8.1 – Scenario C – 4b) End of first FY

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/12/2023	61		Services and other goods	4,500	
		492	Accrued charges		4,500

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+1,125	0	+1,125	0	+1,125
Total impact	**+1,125**	**0**	**+1,125**	**0**	**+1,125**

Exercise 8.1 – Scenario C

4) Record the transactions in the journal and identify the direct and indirect impacts on cash flows for the tenant (also called the lessee), S&L:

   c) During the second financial year (excluding closing operations).

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Exercise 8.1 – Scenario C – 4c) Accruals

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
01/01/2024	492		Accrued charges	4,500	
		61	Services and other goods		4,500

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-1,125	0	-1,125	0	-1,125
Total impact	**-1,125**	**0**	**-1,125**	**0**	**-1,125**

Exercise 8.1 – Scenario C – 4c) Rent payment

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/07/2024	61		Services and other goods	10,800	
		440	Suppliers		10,800
	440		Suppliers	10,800	
		550.0	Credit institutions: Current account		10,800

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	-10,800	0	-10,800	0	-10,800
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+2,700	0	+2,700	0	+2,700
Total impact	**-8,100**	**0**	**-8,100**	**0**	**-8,100**

Exercise 8.1

5) Analyse the differences between the three scenarios above.

Exercise 8.2 – Scenario A

Scenario A (loan with constant amortisation): On 1 August 2023, T&L takes out a loan from a credit institution with constant amortisation of €52,000 per year, a term of five years and an annual interest rate of 4% (payable in arrears).

1) For each of the above scenarios:

    a) Draw up the amortisation table for the loan

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Exercise 8.2 – Qcenario A – 1a) Amortisation table

Loan amount (V)	52,000.00
Loan term (T)	5
Annual interest rate (r)	4%
Constant annual principal payment	€ 10,400.00

→ 52,000.00/5

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+52,000.00			52,000.00
1	01/08/24	-12,480.00	-2,080.00	-10,400.00	41,600.00
2	01/08/25	-12,064.00	-1,664.00	-10,400.00	31,200.00
3	01/08/26	-11,648.00	-1,248.00	-10,400.00	20,800.00
4	01/08/27	-11,232.00	-832.00	-10,400.00	10,400.00
5	01/08/28	-10,816.00	-416.00	-10,400.00	-

$$= -r * SRD_{t-1}$$
$$= -4\% * 52.000,00$$

$= Constant$

Exercise 8.2 – Scenario A

Scenario A (loan with constant amortisation): On 1 August 2023, T&L takes out a loan from a credit institution with constant amortisation of €52,000 per year, a term of five years and an annual interest rate of 4% (payable in arrears).

1) For each of the above scenarios :

   b) Record the journal entries and identify the direct and indirect impacts on cash flows at the start of the contract and at the end of the first financial year.

Exercise 8.2 – Scenario A – 1b) Beginning of the contract

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
01/08/2023	550.0		Credit institutions: Current account	52,000	
		1730	Credit institutions: Current account payable		52,000

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	+52,000	+52,000
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	0	0	0	0	0
Total impact	**0**	**-0**	**0**	**+52,000**	**+52,000**

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Exercise 8.2 – Scenario A – 1b) End of first FY

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/12/2023	6500		Interest, commission and other charges relating to debts	866.67	
		492	Accrued charges		866.67
	1730		Credit institutions: Current account payable	10,400	
		4230	Current portion of amounts payable after more than one year falling due within one year		10,400

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+216.67	0	+216.67	0	+216.67
Total impact	**+216.67**	**0**	**+216.67**	**0**	**+216.67**

Exercise 8.2 – Scenario B

Scenario B (bullet loan): On 1 August 2023, T&L takes out a bullet loan from a financial institution for €52,000, with a term of five years and an annual interest rate of 4% (payable in arrears).

1) For each of the above scenarios:

   a) Draw up the amortisation table for the loan

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Exercise 8.2 – Scenario B – 1a) Amortisation table

Loan amount (V)	52.000,00
Loan term (T)	5
Annual interest rate (r)	4%
Principal payment	€ 52.000.00

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+52,000.00			52,000.00
1	01/08/24	-2,080.00	-2,080.00	0	52,000.00
2	01/08/25	-2,080.00	-2,080.00	0	52,000.00
3	01/08/26	-2,080.00	-2,080.00	0	52,000.00
4	01/08/27	-2,080.00	-2,080.00	0	52,000.00
5	01/08/28	-54,080.00	-2,080.00	-52,000.00	-

$$= -r * SRD_{t-1}$$
$$= -4\% * 52.000,00$$

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Exercise 8.2 – Scenario B

Scenario B (bullet loan): On 1 August 2023, T&L takes out a bullet loan from a financial institution for €52,000, with a term of five years and an annual interest rate of 4% (payable in arrears).

1) For each of the above scenarios :

   b) Record the journal entries and identify the direct and indirect impacts on cash flows at the start of the contract and at the end of the first financial year.

Exercise 8.2 – Scenario B – 1b) Beginning of contract

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
01/08/2023	550.0		Credit institutions: Current account	52,000	
		1730	Credit institutions: Current account payable		52,000

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	+52,000	+52,000
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	0	0	0	0	0
Total impact	**0**	**0**	**0**	**+52,000**	**+52,000**

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Exercise 8.2 – Scenario B – 1b) End of first FY

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/12/2023	6500		Interest, commission and other charges relating to debts	866.67	
		492	Accrued charges		866.67

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	+216.67	0	+216.67	0	+216.67
Total impact	**+216.67**	**-0**	**+216.67**	**0**	**+216.67**

Exercise 8.2

2) Compare the amortisation tables of scenario A of Exercise 8.1 with those of scenarios A and B from exercise 8.2.

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Exercise 8.2 – Loan with constant annuity

Loan amount (V)	52.000,00
Loan term (T)	5
Annual interest rate (r)	4%
Annuity	€ 11 680,61

$$Annuity = \frac{V \times r}{1 - \frac{1}{(1+r)^T}} = \frac{52{,}000 \times 0.04}{1 - \frac{1}{(1+0.04)^5}}$$

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+52,000.00			52,000.00
1	01/08/24	-11,680.61	-2,080.00	-9,600.61	42,399.39
2	01/08/25	-11,680.61	-1,695.98	-9,984.63	32,414.76
3	01/08/26	-11,680.61	-1,296.59	-10,384.02	22,030.74
4	01/08/27	-11,680.61	-881.23	-10,799.38	11,231.36
5	01/08/28	-11,680.61	-449.25	-11,231.36	-

$$= -r * OA_{t-1}$$
$$= -4\% * 52.000{,}00$$

$$= -(Annuity - Interests)$$
$$= -(11.680{,}81 - 2.080{,}00)$$

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Exercise 8.2 – Loan with constant amortisation

Loan amount (V)	52,000.00
Loan term (T)	5
Annual interest rate (r)	4%
Constant annual principal payment	€ 10,400.00

→ 52,000.00/5

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+52,000.00			52,000.00
1	01/08/24	-12,480.00	-2,080.00	-10,400.00	41,600.00
2	01/08/25	-12,064.00	-1,664.00	-10,400.00	31,200.00
3	01/08/26	-11,648.00	-1,248.00	-10,400.00	20,800.00
4	01/08/27	-11,232.00	-832.00	-10,400.00	10,400.00
5	01/08/28	-10,816.00	-416.00	-10,400.00	-

$$= -r * SRD_{t-1}$$
$$= -4\% * 52.000,00$$

$= Constant$

Exercise 8.2 – Bullet loan

Loan amount (V)	52.000,00
Loan term (T)	5
Annual interest rate (r)	4%
Principal payment	€ 52.000.00

n°	Date of reception/payment	Cash flows	Interests	Principal	Outstanding amount
0	01/08/23	+52,000.00			52,000.00
1	01/08/24	-2,080.00	-2,080.00	0	52,000.00
2	01/08/25	-2,080.00	-2,080.00	0	52,000.00
3	01/08/26	-2,080.00	-2,080.00	0	52,000.00
4	01/08/27	-2,080.00	-2,080.00	0	52,000.00
5	01/08/28	-54,080.00	-2,080.00	-52,000.00	-

$$= -r * SRD_{t-1}$$
$$= -4\% * 52.000,00$$

Exercise 8.2 – 2) Comparison

- **8.1. Scenario A (loan with constant annual repayments):** repayments remain the same throughout the term of the loan (the same amount paid each year for 5 years)
- **8.2. Scenario A (loan with equal instalments):** payments decrease over the term of the loan
- **8.2. Scenario B (bullet loan):** small payments throughout the term of the loan (interest only), and a large payment at the end (interest + principal)

→ The cash flow profiles of these various borrowing solutions differ significantly for the borrowing company (as well as for the lender)

APPENDIX

Exercise 8.1 – Scenario B

2) For the scenario described above, record the transaction in the journal and identify the direct and indirect impacts on cash flows for the lessor, whose core business is leasing.

   a) At the beginning of the contract

Exercise 8.1 – Scenario B – 3a) Beginning of the contract

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
01/08/2023	2900		Trade debtors: Customers	46,000	
		26.0	Other tangible fixed assets: Acquisition value		46,000

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	-46,000	+46,000	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	0	0	0	0	0
Total impact	**-46,000**	**+46,000**	**0**	**0**	**0**

Exercise 8.1 – Scenario B

2) For the scenario described above, record the transaction in the journal and identify the direct and indirect impacts on cash flows for the lessor, whose core business is leasing.

   b) At the end of the first FY.

Exercise 8.1 – Scenario B – 3b)End of first FY

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/12/2023	491		Accrued income	766.67	
		743	Miscellaneous operating income		766.67
	400		Customers	8,492.85	
		2900	Trade debtors: Customers		8,492.85

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-191.67	0	-191.67	0	-191.67
Total impact	**-191.67**	**0**	**-191.67**	**0**	**-191.67**

Exercise 8.1 – Scenario B

2) For the scenario described above, record the transaction in the journal and identify the direct and indirect impacts on cash flows for the lessor, whose core business is leasing.

   c) During the second FY.

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Exercise 8.1 – Scenario B – 3c) Accruals

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
01/01/2024	743		Miscellaneous operating income	766.67	
		491	Accrued income		766.67

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	0	0	0	0	0
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	191.67	0	191.67	0	191.67
Total impact	**191.67**	**0**	**191.67**	**0**	**191.67**

Exercise 8.1 – Scenario B – 3c) Annuity reception

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/07/2024	550.0		Credit institutitions: Current account	10,332.85	
		400	Customers		8,492.85
		743	Miscellaneous operating income		1,840.00

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	+10,332.85	0	+10,332.85	0	+10,332.85
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-460	0	-460	0	-460
Total impact	**+9,872.85**	**0**	**+9,872.85**	**0**	**+9,872.85**

Exercise 8.1 – Scenario B

2) For the scenario described above, record the transaction in the journal and identify the direct and indirect impacts on cash flows for the lessor, whose core business is leasing.

   d) At the end of the contract when the option is paid.

Exercise 8.1 – Scenario B – 3d) Option is paid

- Accounting journal

Date	N° SCA Debit	N° SCA Credit	Name	Amount (€) Debit	Amount (€) Credit
31/07/2028	550.0		Credit institutitions: Current account	10,332.85	
		400	Customers		9,935.43
		743	Miscellaneous operating income		397.42
31/07/2028	400		Customers	6,000.00	
		26.0	Other tangible fixed assets: Acquisition value		6,000.00
	550.0		Credit institutitions: Current account	6,000.00	
		400	Customers		6,000.00

Exercise 8.1 – Scenario B – 3d) Option is paid

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	+10,332.85	+6,000	+16,332.85	0	+16,332.85
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-99.35	0	-99.35	0	-99.35
Total impact	**+10,233.50**	**+6,000**	**+16,233.50**	**0**	**+16,233.50**

Exercise 8.1 – Scenario B

2) For the scenario described above, record the transaction in the journal and identify the direct and indirect impacts on cash flows for the lessor, whose core business is leasing.

   e) At the term of the contract when the option is not paid

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Exercise 8.1 – Scenario B – 3e) Option is not paid

- Accounting journal

Date	N° SCA		Name	Amount (€)	
	Debit	Credit		Debit	Credit
31/07/2028	550.0		Etablissements de crédit: Comptes courants	10,332.85	
		400	Customers		9,935.43
		743	Miscellaneous operating income		397.42
31/07/2028	26.0		Other tangible fixed assets: Acquisition value	46,000.00	
		26.9	Other tangible fixed assets: Recorded depreciation		46,000.00

Exercise 8.1 – Scenario B – 3e) Option is not paid

- Impact on cash flows

Timing	CFop (1)	CFinv (2)	FCF (3) = (1) + (2)	CFfin (4)	Δ CASH (5) = (3) + (4)
Direct impact: at accounting entry	+10,332.85	0	+10,332.85	0	+10,332.85
VAT indirect impact: at VAT settlement	0	0	0	0	0
Corporate tax indirect impact: at corporate tex settlement	-99.35	0	-99.35	0	-99.35
Total impact	**+10,233.50**	**0**	**+10,233.50**	**0**	**+10,233.50**